

Development of a biologically inspired artificial lateral line for undulatory robotics

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Introduction

- Undulatory flexible robotics offer an advantage over traditional underwater vehicles in complex environments.
- Researchers have been taking inspiration from fish for undulatory robotics.
- Undulatory vehicles must be able to detect and react to fluid flow rate.
- Sensors traditionally used to measure fluid flow rate cannot be effectively integrated on flexible vehicles.
- Fish have a lateral line system that allows them to sense the flow of the water around them, which could be replicated for underwater vehicles.

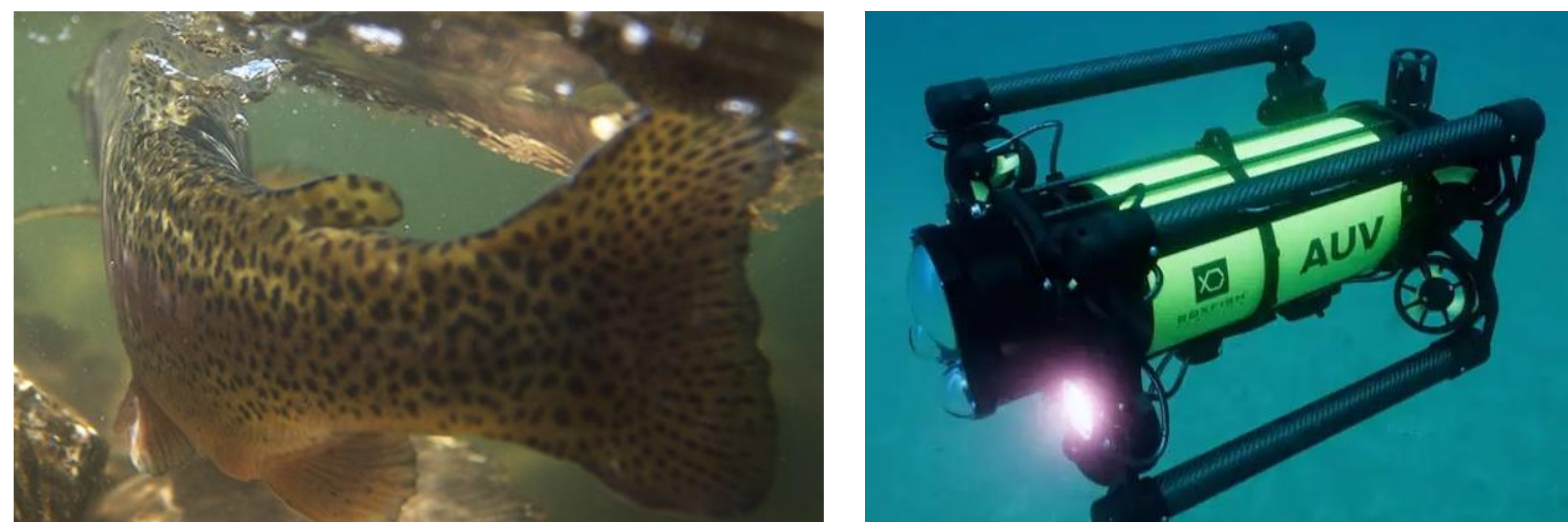


Fig 1- Fish are flexible and able to maneuver through dynamic environments, while underwater vehicles are bulky and often have open propellers that can harm habitats and environments.

Research Goal

Replicate the biological lateral line system using commercial flex sensors and differential pressure sensors.

Methods

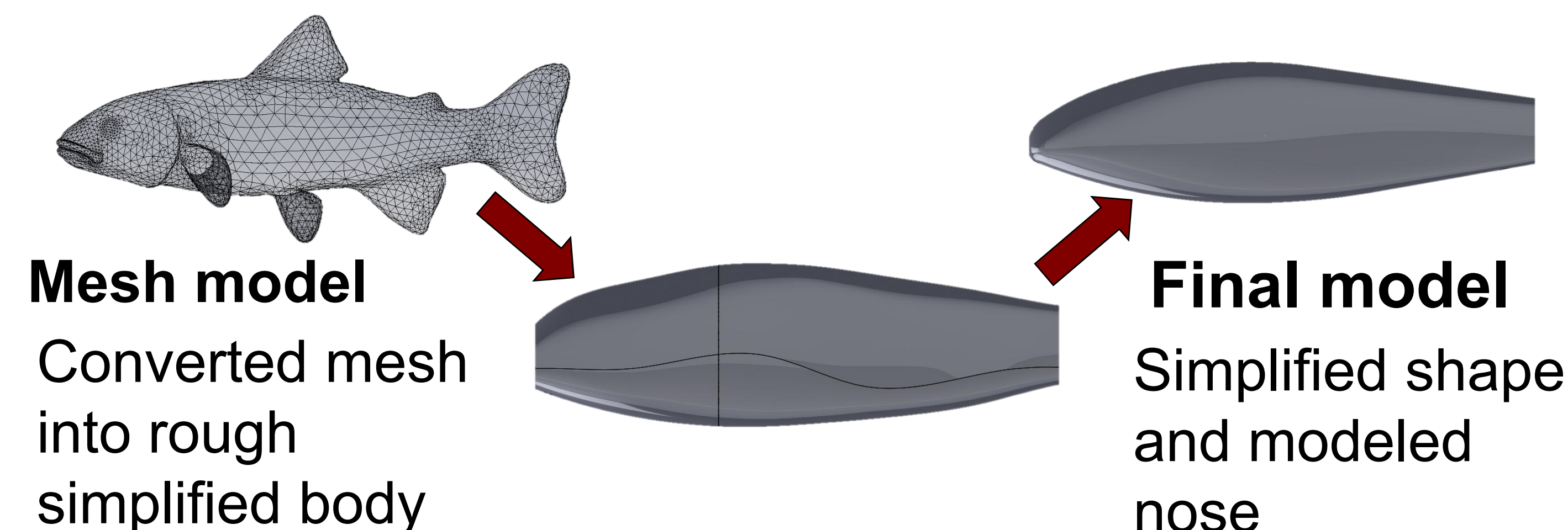


Fig 2- Fish-inspired rig design process.

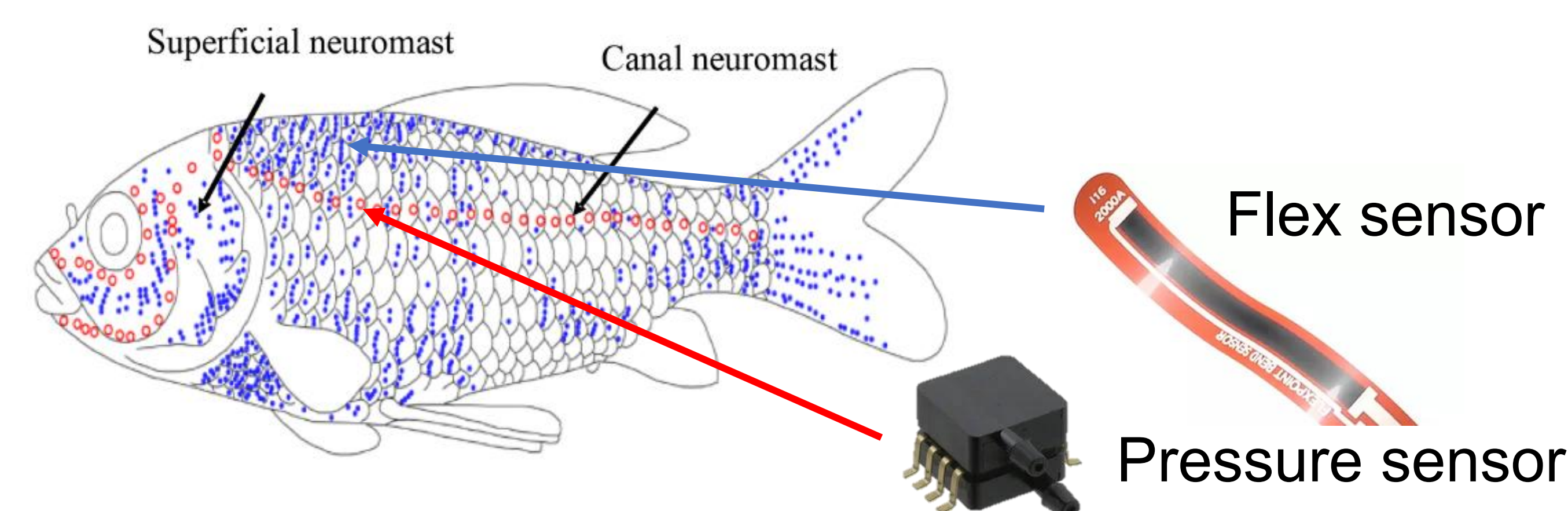


Fig 3- Biologically inspired sensor placement. [1]

1. Design fish-inspired rig for fluid dynamics testing and analysis.
2. Determine optimal sensor placement based on CFD, multiple information theory, and biology.
3. Manufacture rig and install the selected sensors.
4. Test setup in wind tunnel and water tunnel and collect data at set velocities and angles as well as in undulatory motion.



Fig 4- Water tunnel

Results and Future Work

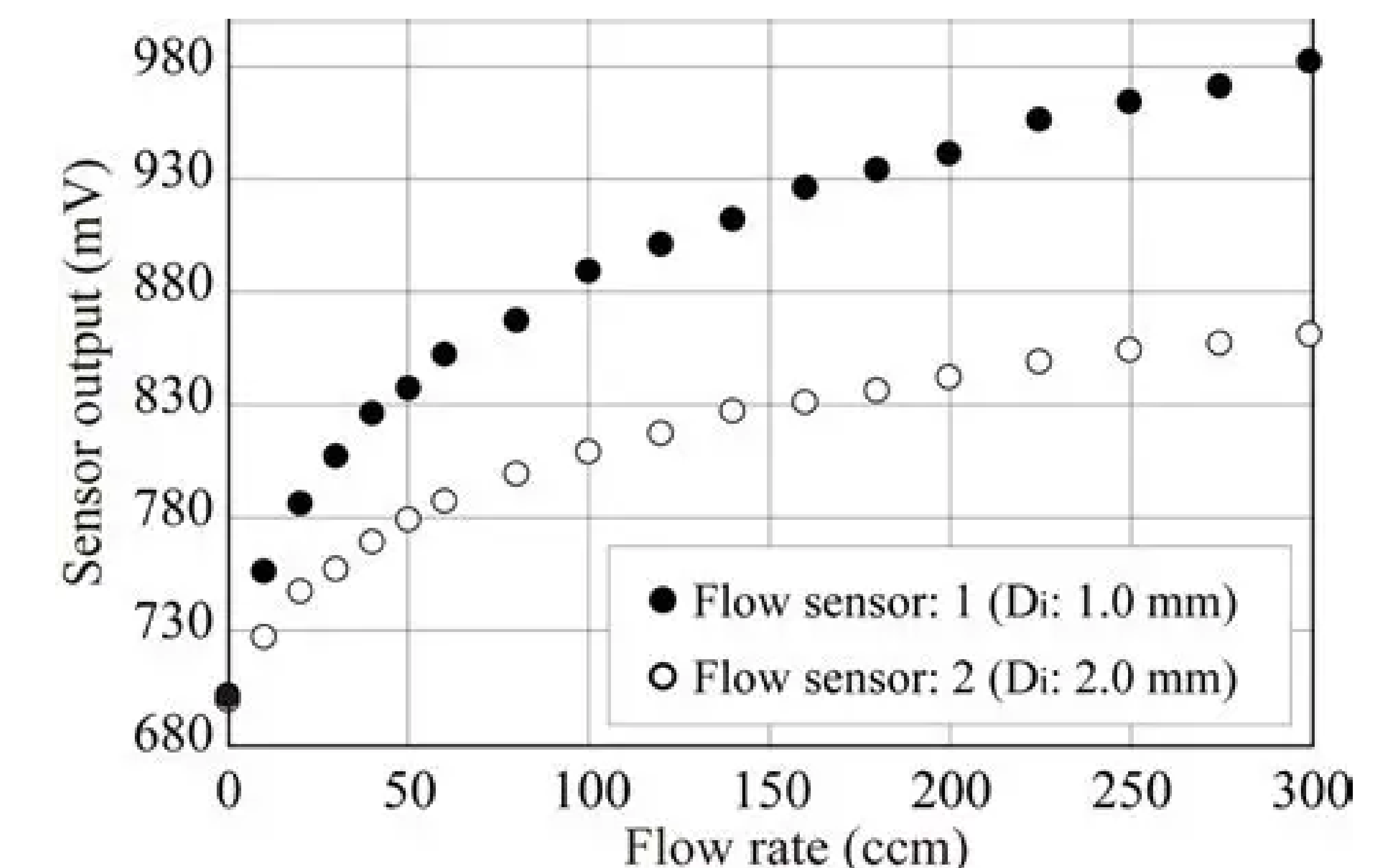


Fig 5- Output from flow sensor at different flow. [2]

- Machine learning model will interpret the sensor data to illustrate a fluid flow field around rig.
- Results will allow for further testing of artificial lateral line system.
- Implement on a biologically inspired flexible robot in the future.

Acknowledgments

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References

- [1] Wu, Jiaxiong, Xuguang Sun, Shudi Yang, Hui Ye, Xiaoce Sun, and Kaifeng Huang. "Flow Field Identification and Water Resistance Calculation Method of Underwater Tracked Vehicle Based on Artificial Lateral Line." *Applied Ocean Research* 142 (January 2024): 103841. <https://doi.org/10.1016/j.apor.2023.103841>.
- [2] Haehnel-Taguchi, Melanie, Otmar Akanyeti, and James C Liao. "Behavior, Electrophysiology, and Robotics Experiments to Study Lateral Line Sensing in Fishes." *Integrative and Comparative Biology* 58, no. 5 (2018): 874–83. <https://doi.org/10.1093/icb/icy066>.